

M I D D E L F A R T R A C I N G

802.1X Network Access Control

Deployment guide for PacketFence, OPNsense and the Cisco access switches at Middelfart (DC1) and Odense (DC2)

Document	Implementation guide
Version	1.1
Date	24 September 2026 (v1.1, corrected after the rollout)
Scope	NW-ASW-DC1-01, NW-ASW-DC2-01 (ports Gi1/0/1-46)
NAC server	SRV-RADIUS-01 · PacketFence 15.2.0 · 10.0.10.16
Status	In production at both sites (pilot completed)

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1 Summary and design

This guide adds 802.1X port authentication, with MAC Authentication Bypass (MAB) as a fallback, to the access ports on both access switches. PacketFence (SRV-RADIUS-01) is the RADIUS server. It decides which VLAN each device gets:

Device / situation	How it authenticates	Result
Domain user or computer	802.1X (PEAP-MSCHAPv2) against AD <code>mfrace.internal</code>	Role default → VLAN 20 Klienter
Unknown or unregistered device	MAB (MAC address)	Role registration → VLAN 555 (captive portal)
Device that breaks policy	802.1X or MAB	Role isolation → VLAN 666 Jail
UniFi access point (one per site, Gi1/0/45)	MAB, MAC registered in PacketFence	Role unifi_ap . PacketFence pushes the interface template <code>PF_UNIFI_AP</code> , which turns the port into a trunk (native 99, tagged 20)
PacketFence unreachable	–	Critical authorization → VLAN 20 (users can keep working)

1.1 Design decisions

- **OPNsense stays the only router.** The DSWs only switch. The registration and jail networks get CARP gateways on the OPNsense HA pairs, just like the existing VLANs.
- **Middelfart is layer 2 to PacketFence.** PacketFence has its own interfaces in VLANs 555 and 666 at DC1 and serves DHCP and DNS there directly.
- **Odense is routed.** The Odense firewall relays DHCP to PacketFence over the existing route-based IPsec tunnel (`ipsec10`).
- **New per-site subnets:** registration `10.0.55.0/24` / `10.10.55.0/24` , jail `10.0.66.0/24` / `10.10.66.0/24` .
- **IBNS 2.0 (new-style) on the ASWs.** It's needed so PacketFence can push an interface template to the AP port. It also makes the port policy clearer.
- **Existing RADIUS group.** The ASWs already use `SRV-RADIUS` → `10.0.10.16` for VTY login. 802.1X reuses this group and leaves login untouched.

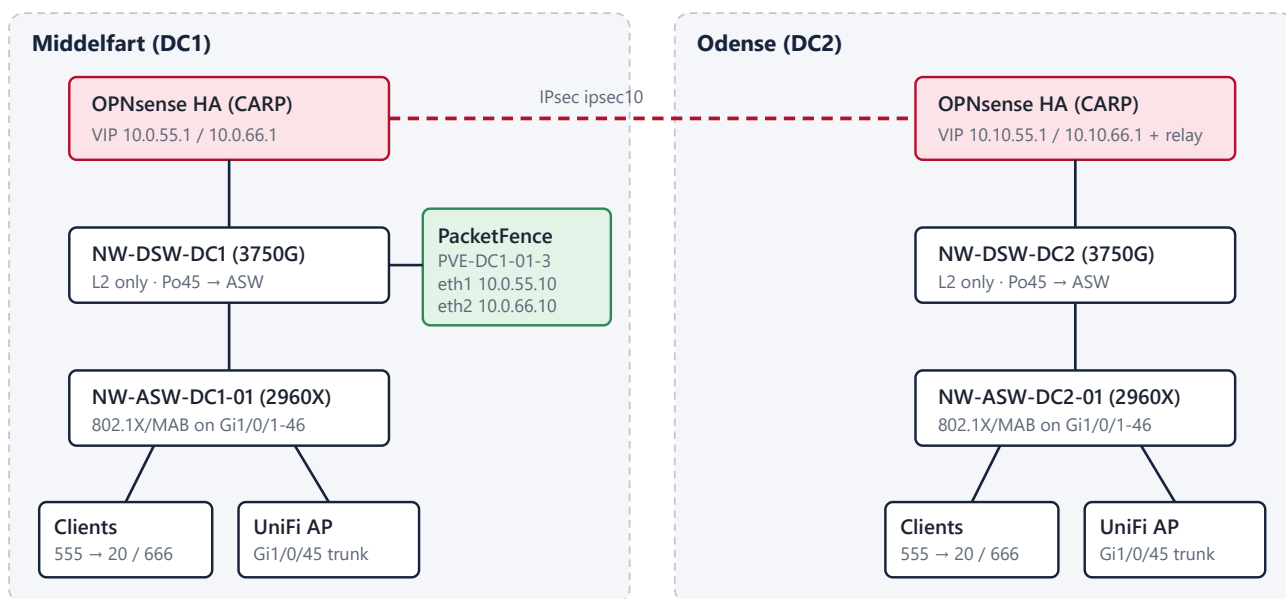


Figure 1 – Target design. RADIUS/CoA from both ASWs goes to 10.0.10.16. Odense DHCP for 555/666 is relayed over IPsec.

2 Current state (audited 18 Sep 2026)

The values in this guide come from the live devices, not only from the repository.

2.1 Devices

Device	Model / software	Relevant finding
NW-ASW-DC1-01 (10.0.99.6)	WS-C2960X-48FPS-L, IOS 15.2(7)E10	VLANs 10, 99 only. Gi1/0/45 = AP trunk (native 99, allowed 10,20,99). No 802.1X yet. LLDP off.
NW-ASW-DC2-01 (10.10.99.6)	WS-C2960X-48FPS-L, IOS 15.2(7)E9	VLAN 99 only. Gi1/0/45 is an access port in VLAN 99 and is <i>notconnect</i> , so the AP isn't plugged in.
NW-DSW-DC1 stack (10.0.99.4)	2× WS-C3750G-48PS-S, 15.0(2)SE9, VTP transparent	VLANs 555/666 exist. They're on the PVE and firewall trunks (Po1/4/7/47/48) but not on Po45 to the ASW .
NW-DSW-DC2 stack (10.10.99.4)	2× WS-C3750G-48TS-S, 15.0(2)SE11, VTP server	VLANs 555/666 exist. Not on Po45 (ASW) or Po47/48 (firewalls) .
OPNsense DC1 (10.0.99.2 / VIP .1)	OPNsense 26.7.3, CARP HA, sync peer 192.168.2.2	lagg0 trunk. Route-based IPsec ipsec10 , with route 10.10.0.0/16 → tunnel. No 555/666 interfaces.
OPNsense DC2 (10.10.99.2 / VIP .1)	OPNsense 26.7.4, CARP HA, sync peer 192.168.2.2	Route 10.0.0.0/16 → ipsec10 . DHCP Relay (Client, MGMT) → SRV-RELAY . No 555/666 interfaces.
SRV-RADIUS-01 (10.0.10.16)	PacketFence 15.2.0, VM 111 on HYP-PVE-DC1-01-3	Joined to AD MFRACE . AD source mfrace (LDAPS). Wrong switch types. Invalid jail interface IP. Default example EAP certificate.

2.2 Problems this guide fixes

#	Problem	Fixed in
1	PacketFence eth2 is 10.66.6.0/24 (the network address)	\$4.1 – new addressing
2	Registration DHCP pool starts at PacketFence's own IP	\$4.2
3	ASW entries in PacketFence are type Cisco::Catalyst_2970	\$4.4 – Cisco IOS v15.0
4	No registration, isolation or default VLAN mapping (built-in defaults are 2/3)	\$4.3
5	VLANs 20/555/666 missing on the ASWs. 555/666 missing on several trunks	\$8, \$9.2
6	RADIUS server certificate is the example certificate (Example Inc.)	\$4.7
7	Sample switch entries 192.168.0.1 and 192.168.1.0/24	\$4.4

3 Addressing plan and NetBox

Site	VLAN	Prefix	Gateway (CARP VIP)	FW node A / B	PacketFence	DHCP pool
Middelfart	555 Registration	10.0.55.0/24	10.0.55.1	.2 / .3	10.0.55.10 (eth1)	.20 – .250
Middelfart	666 Jail	10.0.66.0/24	10.0.66.1	.2 / .3	10.0.66.10 (eth2)	.20 – .250
Odense	555 Registration	10.10.55.0/24	10.10.55.1	.2 / .3	relayed → 10.0.55.10	.20 – .250
Odense	666 Jail	10.10.66.0/24	10.10.66.1	.2 / .3	relayed → 10.0.66.10	.20 – .250

Reserved addresses: .1 is the CARP VIP, .2/.3 are the firewall nodes, .10 is PacketFence, and .11–.19 are spare.

3.1 NetBox (10.0.10.21)

1. Delete the prefixes 10.55.5.0/24 and 10.66.6.0/24 .
2. Create the four prefixes above. Link them to their site (Middelfart / Odense) and VLAN (555 / 666 for that site).
3. Create IP addresses for the gateways (.1, role *CARP*), firewall nodes (.2/.3) and PacketFence (.10). Mark the DHCP ranges as *IP Ranges*.
4. Fix the typo in the VLAN name: *Vlan 555 - Registraion* → *Registration*.

4 PacketFence configuration

Web admin: `https://10.0.10.16:1443`. Before you start, back up the configuration from the shell:

```
tar czf /root/pf-conf-$(date +%F).tgz /usr/local/pf/conf
```

4.1 Interfaces

Configuration → Network Configuration → Interfaces

Interface	Proxmox tag	Old IP	New IP	Type	Enforcement
eth0	10	10.0.10.16/24	<i>unchanged</i>	Management	–
eth1	555	10.55.5.10/24	10.0.55.10/24	Registration	VLAN
eth2	666	10.66.6.0/24	10.0.66.10/24	Isolation	VLAN

4.2 Networks

Configuration → Network Configuration → Networks. Delete `10.55.5.0` and `10.66.6.0`, then create these four:

Network	Type	Gateway	DNS	Next hop	DHCP range
10.0.55.0/24	Registration	10.0.55.1	10.0.55.10	–	10.0.55.20 – 10.0.55.250
10.0.66.0/24	Isolation	10.0.66.1	10.0.66.10	–	10.0.66.20 – 10.0.66.250
10.10.55.0/24	Registration (routed)	10.10.55.1	10.0.55.10	10.0.55.1	10.10.55.20 – 10.10.55.250
10.10.66.0/24	Isolation (routed)	10.10.66.1	10.0.66.10	10.0.66.1	10.10.66.20 – 10.10.66.250

Gateway on the Middelfart networks: PacketFence always uses its own interface IP as the clients' gateway on a layer 2 network, so 10.0.55.10 and 10.0.66.10. The field is not shown in the UI and cannot be set to the firewall VIP. Only the routed Odense networks take a gateway (the OPNsense VIP) and a next hop.

For all four, enable *DHCP server* and *DNS*, set the lease time to 30 seconds, and leave NAT disabled. The next hop is a CARP VIP, so PacketFence's return routes survive a firewall failover.

4.3 Roles and the switch group

Configuration → Policies and Access Control → Roles: create the role `unifi_ap` (*UniFi access points*). Use an underscore, not a hyphen, because the name becomes part of a configuration key.

... → *Network Devices → Switch Groups → middelfart-racing*:

Tab / setting	Value
Definition → Type	Cisco IOS v15.0
Definition → Mode	Production (individual switches override this to <i>Testing</i> during the pilot)
Definition → Deauthentication method	RADIUS
Definition → Use CoA	Enabled
Roles → Role by VLAN ID	Enabled
registration / isolation	555 / 666
default	20
unifi_ap	99
RADIUS → Secret passphrase	<RADIUS-SECRET> (unchanged, already set)
SNMP → Version / Community read	v2c / <SNMP-RO>

4.4 Switches

... → *Network Devices* → *Switches*. Edit the existing entries. Don't create new ones.

Switch	Description	Type	Mode
10.0.99.6	NW-ASW-DC1-01	Cisco IOS v15.0 (was Catalyst_2970)	Testing → Production after the pilot
10.10.99.6	NW-ASW-DC2-01	Cisco IOS v15.0 (was Catalyst_2970)	Testing → Production after the pilot
10.0.99.7 / 10.10.99.7	NW-OOB-DC*-01	Cisco IOS v15.0 (was Catalyst_2970). No 802.1X on these.	unchanged
192.168.0.1, 192.168.1.0/24	PacketFence sample entries	Delete	

If CoA doesn't work

If VLAN changes after registration don't take effect during the pilot, change the ASW type to **Cisco IOS v15.5** and test again. Both modules support Catalyst 2960 switches.

4.5 Active Directory source and 802.1X

The AD source `mfrace` already exists (LDAPS, `srv-adds-01/02.mfrace.internal`, base DN `DC=mfrace,DC=internal`), and PacketFence is joined to the domain `MFRACE`. Add or check the authentication rule:

Rule (class: authentication)	Condition	Actions
<code>catchall</code> in source <code>mfrace</code>	match all	<code>Role = default</code> , <i>Access duration</i> = 1 day (reauthentication renews it)

Realms (required). Windows sends the user as `MFRACE\user`. PacketFence only strips the domain for realms it knows, so without these the AD lookup searches for `sAMAccountName=MFRACE\user`, matches nothing and the login is rejected. Under *Policies and Access Control* → *Realms* create `MFRACE` and `mfrace.internal`, both with domain `mfrace` and *Strip on RADIUS authorization* enabled, and list them as the source's associated realms.

Computer authentication. A computer authenticates as `host/PC.mfrace.internal`, which no AD account carries as its name. Add `dnsHostName` to *Search Attributes* on the `mfrace` source, otherwise no rule matches and every PC is rejected before a user logs in.

Then create a connection profile under *Policies and Access Control* → *Connection Profiles*:

Setting	Value
Profile name	wired-dot1x
Filter	Connection Type = Ethernet-EAP
Sources	mfrace
Automatically register devices	Enabled

The default profile stays in place for MAB and unregistered devices, which get the captive portal. Remove the Email, SMS and Sponsor sources from the default profile unless you want guest self-registration.

4.6 UniFi AP: node and RADIUS filter

1. *Nodes* → *Create*: MAC `9c:05:d6:ba:c1:c9` (Middelfart AP, seen on NW-ASW-DC1-01 Gi1/0/45). Status *registered*, role `unifi_ap`, unregistration date *none*, notes "UniFi AP DC1".
2. Look up the MAC of the Odense AP in the UniFi controller (SRV-UNIFI-01) and register it the same way. It isn't connected today.
3. *Configuration* → *Advanced Access Configuration* → *Filter Engines* → *RADIUS Filters*: add the filter below. Or add it to `/usr/local/pf/conf/radius_filters.conf` and run `pfcmd configreload hard`.

```
[UNIFI_AP_TEMPLATE]
status=enabled
description=Push the PF_UNIFI_AP interface template to UniFi AP ports
top_op=and
condition=node_info.category == "unifi_ap"
scopes=returnRadiusAccessAccept
merge_answer=yes
answer.0=Cisco-AVPair = interface-template-name=PF_UNIFI_AP
```

Security note

MAB only checks a MAC address, and MAC addresses can be spoofed. Anyone who copies the AP MAC and plugs into an access port gets a trunk to VLANs 20 and 99. Keep the AP ports physically secured. If that's not good enough, use the static fallback in §10.4 and restrict it with port security.

4.7 RADIUS server certificate

PacketFence still uses the example certificate (*CN=Example Server Certificate*). Windows clients that validate the server certificate won't connect with it, so replace it before the pilot:

1. Request a certificate from the AD CS CA (SRV-PKI-02 is running. SRV-PKI-01 is currently stopped).
Subject/SAN: `srv-radius-01.mfrace.internal` . EKU: *Server Authentication*.
2. Upload it under *Configuration* → *System Configuration* → *SSL Certificates* → *RADIUS* together with the private key and the CA chain.
3. Restart RADIUS: `/usr/local/pf/bin/pfcmd service radiusd restart` .

4.8 Apply

```
/usr/local/pf/bin/pfcmd configreload hard
/usr/local/pf/bin/pfcmd service pf restart
/usr/local/pf/bin/pfcmd service pf status | grep -Ev "started|disabled" # should print nothing
```

5 Proxmox (verification only)

PacketFence runs as **VM 111 SRV-RADIUS-01** on **HYP-PVE-DC1-01-3** (10.0.99.14). Its NICs are already tagged correctly on the VLAN-aware bridge `vmbr0`, and the host trunk (DSW Po7) already carries 555/666. **NO CHANGE**

```
net0: virtio=BC:24:11:60:38:67,bridge=vmbr0,tag=10      # eth0 management
net1: virtio=BC:24:11:6F:4B:6D,bridge=vmbr0,tag=555    # eth1 registration
net2: virtio=BC:24:11:21:53:79,bridge=vmbr0,tag=666    # eth2 isolation
```

HA / migration

All three DC1 hosts have the same `vmbr0` (VLAN-aware, bridge-vids 2-4094), and all three host trunks (Po1, Po4, Po7) carry 555/666. The VM can live-migrate without any network changes.

6 OPNsense – Middelfart (DC1)

HA pair NW-FW-EDGE-DC1-01 (MASTER, .2) and NW-FW-EDGE-DC1-02 (.3). The DSW trunks to both firewalls (Po47, Po48) already carry 555/666.

6.1 VLAN devices and interfaces (on both nodes)

OPNsense HA sync does **not** sync VLAN devices or interface assignments. Do steps 1–2 on each node.

1. *Interfaces* → *Devices* → *VLAN*: add `lagg0 tag 555` (description *REG*) and `lagg0 tag 666` (*JAIL*).
2. *Interfaces* → *Assignments*: assign both, enable them, and set static IPv4:

Interface	Device	NW-FW-EDGE-DC1-01	NW-FW-EDGE-DC1-02
REG	lagg0_vlan555	10.0.55.2/24	10.0.55.3/24
JAIL	lagg0_vlan666	10.0.66.2/24	10.0.66.3/24

6.2 CARP VIPs, aliases and rules (on the master, then sync)

Interfaces → *Virtual IPs* → *Settings*:

Mode	Interface	Address	VHID	Description
CARP	REG	10.0.55.1/24	55	REG gateway
CARP	JAIL	10.0.66.1/24	66	JAIL gateway

Firewall → *Aliases*: `PF_REG` = 10.0.55.10, `PF_JAIL` = 10.0.66.10.

Firewall → *Rules* → *REG* (JAIL is identical with `JAIL net` / `PF_JAIL`):

#	Action	Proto	Source	Destination	Port	Description
1	Pass	UDP	REG net	PF_REG	67	DHCP (unicast renew)
2	Pass	TCP/UDP	REG net	PF_REG	53	DNS
3	Pass	TCP	REG net	PF_REG	80, 443	Captive portal
4	Block + log	any	REG net	any	any	Everything else

Don't enable DHCP here

PacketFence serves DHCP directly in VLANs 555/666 at DC1. Don't enable Kea or dnsmasq on REG/JAIL, and don't add them to DHCP Relay.

6.3 What doesn't change at DC1

Item	Why no change is needed
IPsec	Route-based. The existing route 10.10.0.0/16 → ipsec10 already covers 10.10.55/66, and ipsec10 passes all inbound traffic.
Outbound NAT	Only alias NAT_Primary is NATed. 10.0.55/66 aren't in it, so registration and jail clients get no internet access.
RADIUS / CoA / SNMP	MGMT and SRV pass all inbound traffic, so switch ↔ 10.0.10.16 on UDP 1812/1813/3799/161 already works.
DHCRelay SRV-RELAY	Already sends Client and MGMT DHCP to 10.0.10.16 as well, which PacketFence uses for device fingerprinting.

Finally, go to *System* → *High Availability* → *Status* and click **Synchronize and reconfigure all**. On the backup node, check that the VIPs show as *BACKUP*.

7 OPNsense – Odense (DC2)

HA pair NW-FW-EDGE-DC2-01 (MASTER, .2) and NW-FW-EDGE-DC2-02 (.3). Follow the same steps as §6.1 and §6.2, with these values:

Interface	Device	Node 01	Node 02	CARP VIP (VHID)
REG	lagg0_vlan555	10.10.55.2/24	10.10.55.3/24	10.10.55.1 (55)
JAIL	lagg0_vlan666	10.10.66.2/24	10.10.66.3/24	10.10.66.1 (66)

Create the aliases `PF_REG` = 10.0.55.10 and `PF_JAIL` = 10.0.66.10. Use the same four rules per interface as §6.2, with rule 1 changed to allow DHCP to the firewall for the relay:

#	Action	Proto	Source	Destination	Port	Description
1	Pass	UDP	any	any	67	DHCP to the relay
2–4	As in §6.2 (DNS and portal to <code>PF_REG</code> / <code>PF_JAIL</code> , then block + log)					

7.1 DHCP relay

Services → *DHCRelay*. Create two **new destinations**. Don't reuse `SRV-RELAY` , because it also contains 10.0.10.12 and 10.0.10.13, and those DHCP servers would answer registration clients too.

Destination	Server
PF-REG	10.0.55.10
PF-JAIL	10.0.66.10

Relay interface	Destination	CARP depend on	Agent info
REG	PF-REG	VIP 10.10.55.1	off
JAIL	PF-JAIL	VIP 10.10.66.1	off

Setting *CARP depend on* matches your existing relays and makes only the MASTER relay, so PacketFence never sees duplicate requests.

7.2 What doesn't change at DC2

- **Routing:** the route 10.0.0.0/16 → `ipsec10` already exists, and `ipsec10` passes all inbound traffic.
- **RADIUS and CoA:** MGMT passes all inbound traffic, and PacketFence's CoA to 10.10.99.6 arrives over `ipsec10` .
- **NAT:** 10.10.55/66 aren't in `NAT_Primary` , so there's no internet access for these networks.

Synchronize to the backup node as in §6.3.

8 Distribution switches (DSW)

The DSWs only switch (`ip default-gateway` , no `ip routing`). They need no SVIs, ACLs or helper addresses. The only change is to allow VLANs 555/666 on the trunks that are missing them.

8.1 NW-DSW-DC1 (10.0.99.4)

```
! 555/666 already on Po1/Po4/Po7 (PVE) and Po47/Po48 (firewalls)
configure terminal
interface Port-channel45
  description "NW-ASW-DC1-01 | Port g1/0/47-48"
  switchport trunk allowed vlan add 555,666
end
write memory
```

8.2 NW-DSW-DC2 (10.10.99.4)

```
! VLANs 555/666 already exist (VTP server). Only the trunks are missing them.
configure terminal
interface range Port-channel45 , Port-channel47 - 48
  switchport trunk allowed vlan add 555,666
end
write memory
```

8.3 Verify

```
show interfaces trunk | include Po45|Po47|Po48
show spanning-tree vlan 555,666 | include Po
```

9 Access switches (ASW): 802.1X + MAB

Apply the same configuration to **NW-ASW-DC1-01** and **NW-ASW-DC2-01**. Start at DC1.

Before you start

- Connect a console cable, or schedule `reload in 20`, and cancel it with `reload cancel` once everything works.
- Back up the config: `copy running-config flash:pre-dot1x.cfg`.
- `authentication convert-to new-style` is **one-way**. You can only get back to legacy mode by restoring the backup config and reloading (\$14).
- Identify the devices that are online today in VLAN 99: DC1 Gi1/0/37, 38, 46 and DC2 Gi1/0/46. Once 802.1X is on, they'll land in VLAN 20 or 555 instead. If they need management access, exclude those ports (\$9.5).

9.1 Switch to IBNS 2.0

```
configure terminal
authentication convert-to new-style
! answer "yes" to the prompt
```

9.2 VLANs, uplink and global AAA

```
! --- VLANs (the ASWs are VTP transparent, so VLANs must exist locally) ---
vlan 20
  name Klienter
vlan 99
  name Management
vlan 555
  name Registration
vlan 666
  name Jail
vlan 999
  name Blackhole
!
interface Port-channel24
  switchport trunk allowed vlan add 555,666
!
! --- 802.1X / RADIUS (reuses the existing group SRV-RADIUS) ---
dot1x system-auth-control
aaa authentication dot1x default group SRV-RADIUS
aaa authorization network default group SRV-RADIUS
aaa accounting identity default start-stop group SRV-RADIUS
!
radius server SRV-RADIUS-01
  automate-tester username pf-probe ignore-acct-port idle-time 3
radius-server dead-criteria time 5 tries 3
radius-server deadtime 15
radius-server vsa send authentication
radius-server vsa send accounting
!
aaa server radius dynamic-author
  client 10.0.10.16 server-key <RADIUS-SECRET>
  auth-type any
  port 3799
!
snmp-server community <SNMP-RO> RO
lldp run
```

Leave these as they are

The existing key 7 ... under radius server SRV-RADIUS-01, and the login lists VTY_AUTHEN / VTY_AUTHOR. Never add aaa authentication login default group ..., or you'll be locked out of the console when PacketFence is down.

9.3 Control policy (IBNS 2.0)

This policy tries 802.1X first. If the device doesn't answer or fails, it falls back to MAB, so PacketFence can send it to registration. If PacketFence is unreachable, the port is authorized in VLAN 20.

```

service-template PF_CRITICAL
vlan 20
!
class-map type control subscriber match-all DOT1X_NO_RESP
match method dot1x
match result-type method dot1x agent-not-found
class-map type control subscriber match-all DOT1X_FAILED
match method dot1x
match result-type method dot1x authoritative
class-map type control subscriber match-all MAB_FAILED
match method mab
match result-type method mab authoritative
class-map type control subscriber match-all AAA_DOWN
match result-type aaa-timeout
match authorization-status unauthorized
!
policy-map type control subscriber PF-DOT1X-MAB
event session-started match-all
  10 class always do-until-failure
    10 authenticate using dot1x priority 10
event authentication-failure match-first
  5 class AAA_DOWN do-until-failure
    10 activate service-template PF_CRITICAL
    20 authorize
    30 pause reauthentication
  10 class DOT1X_NO_RESP do-until-failure
    10 terminate dot1x
    20 authenticate using mab priority 20
  20 class DOT1X_FAILED do-until-failure
    10 terminate dot1x
    20 authenticate using mab priority 20
  30 class MAB_FAILED do-until-failure
    10 terminate mab
    20 authentication-restart 60
  40 class always do-until-failure
    10 terminate dot1x
    20 terminate mab
    30 authentication-restart 60
event agent-found match-all
  10 class always do-until-failure
    10 terminate mab
    20 authenticate using dot1x priority 10
event aaa-available match-all
  10 class always do-until-failure
    10 clear-session
event violation match-all
  10 class always do-until-failure
    10 replace
event inactivity-timeout match-all
  10 class always do-until-failure
    10 clear-session

```

9.4 Access port template

The port settings live in an **interface template** rather than directly on the interface. Settings written directly on an interface override a template, so this is what lets PacketFence replace the whole port config for the AP (\$10).

```

template PF_ACCESS
description 802.1X/MAB - PacketFence
switchport mode access
switchport access vlan 555
spanning-tree portfast
spanning-tree bpduguard enable
dot1x pae authenticator
dot1x timeout tx-period 3
dot1x timeout quiet-period 2
mab
access-session host-mode single-host
access-session closed
access-session port-control auto
authentication periodic
authentication timer reauthenticate server
service-policy type control subscriber PF-DOT1X-MAB

```

9.5 Apply to the ports

Remove the old static settings first. Otherwise they override the template.

```

! --- PILOT: one free port first (Gi1/0/1) ---
default interface GigabitEthernet1/0/1
interface GigabitEthernet1/0/1
source template PF_ACCESS
access-session interface-template sticky timer 30
no snmp trap link-status
end

! --- ROLLOUT (after the pilot in §12 passes) ---
! Never include Gi1/0/47-48 or Po24 (uplink to the DSW).
! Leave out any port that must stay in VLAN 99 (see the warning at the top of §9).
configure terminal
default interface range GigabitEthernet1/0/1 - 46
interface range GigabitEthernet1/0/1 - 46
source template PF_ACCESS
access-session interface-template sticky timer 30
no snmp trap link-status
end
write memory

```

Gi1/0/45 (AP port) gets the same config

That's intended. The AP authenticates with MAB like any other device, and PacketFence then switches the port to the AP template (§10). Only use `default interface Gi1/0/45` during a maintenance window, because the AP's clients drop for about a minute.

10 Automatic AP port configuration

Goal: when a UniFi AP is plugged into any 802.1X port (normally Gi1/0/45), the port automatically becomes a trunk with management untagged on VLAN 99 and wireless clients tagged on VLAN 20. This matches today's manual config on NW-ASW-DC1-01 Gi1/0/45.

10.1 How it works

1. The port starts with template `PF_ACCESS`. The UniFi AP doesn't speak 802.1X, so after about 10 seconds the switch falls back to MAB.
2. PacketFence recognises the MAC as a registered node in role `unifi_ap`. The RADIUS filter `UNIFI_AP_TEMPLATE` (\$4.6) adds `Cisco-AVPair = interface-template-name=PF_UNIFI_AP` to the Access-Accept.
3. The switch binds the dynamic template `PF_UNIFI_AP` to the port. It does **not** replace `PF_ACCESS`: the two are merged, and the dynamic template wins wherever both set the same command. That is why `PF_UNIFI_AP` has to set `access-session host-mode multi-host`. With the `single-host` from `PF_ACCESS` still in force, the first Wi-Fi client behind the AP counts as a second MAC, triggers a security violation and the trunk never comes up.
4. Do not use `access-session port-control force-authorized` in the AP template. It ends the authentication session, the session was what bound the dynamic template, so the port falls back to `PF_ACCESS` and the whole thing loops every 40 seconds.
5. The port itself must carry no `switchport` commands. Settings written on the interface beat any template, so run `default interface` before `source template PF_ACCESS`. A leftover `switchport mode access` keeps the port an access port for ever.
6. `access-session interface-template sticky` keeps the template bound while the link stays up. If the link goes down (AP unplugged), the port returns to `PF_ACCESS` after 30 seconds, so a different device in that port goes through 802.1X/MAB again.

10.2 Template on both ASWs

```
template PF_UNIFI_AP
description UniFi AP - pushed by PacketFence
switchport mode trunk
switchport trunk native vlan 99
switchport trunk allowed vlan 20,99,100
access-session host-mode multi-host
access-session port-control auto
spanning-tree portfast edge trunk
spanning-tree bpduguard enable
```

Today's DC1 AP port also allows VLAN 10. Only add it (`allowed vlan 10,20,99`) if an SSID really uses VLAN 10. By default, servers shouldn't be reachable over Wi-Fi.

10.3 Verify

```
show access-session interface Gi1/0/45 details      ! Method: mab, Status: Authorized
show template binding target Gi1/0/45              ! PF_UNIFI_AP dynamic
show interfaces Gi1/0/45 switchport | include Mode|Native|Trunking VLANs
show mac address-table interface Gi1/0/45          ! AP on 99, Wi-Fi clients on 20
```

In PacketFence's *Auditing* → *RADIUS Audit Logs*, the Access-Accept for `9c:05:d6:ba:c1:c9` must contain the `interface-template-name` attribute.

10.4 Fallback: static AP port

If the pilot shows the dynamic template doesn't bind on your IOS release, configure the AP port statically and leave it out of 802.1X. Port security limits it to the AP's MAC:

```
default interface GigabitEthernet1/0/45
interface GigabitEthernet1/0/45
  description UniFi AP (static, excluded from NAC)
  switchport mode trunk
  switchport trunk native vlan 99
  switchport trunk allowed vlan 20,99
  switchport port-security maximum 1 vlan access
  switchport port-security mac-address sticky
  switchport port-security violation restrict
  switchport port-security
  spanning-tree portfast trunk
```

With port security on a trunk, the AP's own MAC is learned on the native VLAN. Wireless clients (VLAN 20) aren't limited. If you use this fallback, test it with the AP connected before you rely on it.

11 Windows clients (Group Policy)

Users sign in with 802.1X (PEAP-MSCHAPv2) using their MFRACE AD credentials. Deploy the settings with a GPO linked to the OUs that contain the computers:

Setting	Value
System Services → <i>Wired AutoConfig</i> (dot3svc)	Automatic
Security Settings → <i>Wired Network (IEEE 802.3)</i>	New policy MFRACE-Wired , Use Windows Wired Auto Config enabled Policies
Network authentication method	Microsoft: Protected EAP (PEAP)
Verify the server's identity with its certificate	Enabled. Connect to servers: <code>srv-radius-01.mfrace.internal</code>
Trusted Root CA	Your AD CS root CA (the CA that issued the certificate in §4.7)
Inner method	Secured password (EAP-MSCHAP v2), Automatically use my Windows logon name and password
Authentication mode	User or computer authentication

Why "user or computer"

The computer authenticates at boot, so domain logon, GPO processing and login scripts work before a user signs in. Once the user signs in, the connection re-authenticates as that user. PacketFence gives both the `default` role (VLAN 20).

Fallback when PacketFence is unreachable

Without it the critical VLAN (§9.3) does not help: the switch authorizes the port in VLAN 20, but Windows sees *Authentication failed* and blocks the connection itself, so the PC has an address and still reaches nothing. The setting is called **Fallback to unauthorized network access** (`OneXEnforced=false` in the profile). The GPO wired policy does **not** expose it: it only exists on the adapter's Authentication tab and in a LAN profile XML. Deploy it as a LAN profile instead: build the profile on one PC, export it with `netsh lan export profile`, and apply it at boot with a startup script (`netsh lan add profile filename=... interface=...`), or use `docs/wired-dot1x/New-MFRACEWiredProfile.ps1` from the repository, which builds the profile from the PC's own certificate store. A GPO wired policy always overrides local profiles, so remove the policy if you deploy profiles this way. Run the script elevated (SYSTEM in a startup script), otherwise you get a user profile and the machine profile needed for authentication before logon is missing.

EAP-TLS or PEAP

Both work against this PacketFence. PEAP-MSCHAPv2 needs no client certificate, but Credential Guard (on by default in Windows 11 Enterprise/Education 22H2+) blocks single sign-on with it, and the PC then quietly falls back to MAB. EAP-TLS with a computer certificate from MFRACE-Issuing-CA avoids that and sends no passwords. Whichever you choose, use the same one everywhere, and check `sc query dot3svc`: with the Wired AutoConfig service stopped the PC never answers EAPOL at all and lands on MAB.

Devices that can't do 802.1X, such as printers, register through the captive portal, or an admin registers them by MAC in PacketFence with the role they need.

12 Pilot, verification and rollout

12.1 Order of work

#	Step	Section	Risk
1	NetBox prefixes	§3	none
2	PacketFence: interfaces, networks, roles, switches (ASWs in Testing), AP node, RADIUS filter, certificate	§4	low
3	OPNsense DC1 + DC2 (both nodes each)	§6–7	low
4	DSW trunks	§8	low
5	Test VLAN 555 by hand: put a PC on an ASW port set to <code>switchport access vlan 555</code> . It must get 10.0.55.20+ and the portal.	–	none
6	ASW DC1: global config (§9.1–9.4) + pilot port Gi1/0/1	§9	medium
7	Pilot tests (§12.2) with the switch in PacketFence mode <i>Testing</i> , then <i>Production</i>	§12.2	low
8	AP test on Gi1/0/45 (maintenance window)	§10	medium
9	Rollout Gi1/0/1–46 on DC1	§9.5	medium
10	Repeat steps 5–9 at DC2 (10.10.55.x via relay)	–	medium

12.2 Pilot tests

✓	Test	Expected result
☐	test aaa group SRV-RADIUS <user> <pass> new-code	User successfully authenticated
☐	Domain PC with the GPO applied, plugged into Gi1/0/1	Method dot1x, Authorized, VLAN 20, IP 10.0.20.x
☐	Unknown device (no 802.1X)	Method mab, VLAN 555, IP 10.0.55.20+, browser opens the portal
☐	Register the device in PacketFence (or via the portal)	CoA bounces the port. The device lands in VLAN 20 within about 10 seconds
☐	Put the node in isolation (<i>Nodes</i> → <i>Apply security event</i>)	CoA → VLAN 666, IP 10.0.66.20+, only the portal reachable
☐	Stop RADIUS on PacketFence for 2 minutes, then reconnect a PC	Port authorized by PF_CRITICAL in VLAN 20. Re-authenticates when RADIUS is back
☐	SSH to the switch as adm-jukn	Login still works (VTY lists unchanged)
☐	UniFi AP on Gi1/0/45	Template PF_UNIFI_AP bound, AP online in UniFi, Wi-Fi client gets 10.0.20.x
☐	DC2: unknown device	IP 10.10.55.20+ through the OPNsense relay, portal loads from 10.0.55.10

12.3 Useful commands

```
show access-session                                ! all sessions
show access-session interface Gi1/0/1 details
show access-session interface Gi1/0/1 policy
show aaa servers | include SRV-RADIUS|State|Dead
show template binding target Gi1/0/45
debug access-session all                          ! plus: terminal monitor. Turn off with "undebg all"
```

13 Troubleshooting

Symptom	Likely cause	Check / fix
No RADIUS request reaches PacketFence	Switch not in <code>switches.conf</code> , or wrong secret	PacketFence <i>RADIUS Audit Log</i> . Check that the source IP is 10.0.99.6 or 10.10.99.6 and that the secret matches.
Device stays in VLAN 555 after registering	CoA blocked or wrong switch type	UDP 3799 from 10.0.10.16. Check <code>aaa server radius dynamic-author</code> and the <code>server-key</code> . Try type IOS v15.5.
Device in 555 gets no IP	VLAN missing on a trunk, or PacketFence DHCP not listening	<code>show interfaces trunk</code> on the ASW and DSW. <code>pfcmd service pfdhcp status</code> . At DC2, check the relay and firewall rule 1.
802.1X fails and the device ends up in 555	Certificate not trusted, or wrong server name in the GPO	Windows event log <i>Wired-AutoConfig</i> . PacketFence <i>RADIUS Audit Log</i> → EAP reason.
802.1X fails with an AD error	Domain join / ntlm_auth	<i>Configuration</i> → <i>Policies</i> → <i>Domains</i> → <i>Test join</i> . Check time sync with <code>srv-adds-01/02</code> .
AP port stays an access port in VLAN 99	Template not returned or not bound	Access-Accept in the audit log. <code>show template binding</code> . Is there interface config overriding the template (§9.5 default interface)?
AP port flaps between trunk and access	<code>sticky</code> missing	<code>access-session interface-template sticky timer 30</code> on the interface.
Everyone in VLAN 20 while PacketFence is healthy	Switch marked RADIUS as dead	<code>show aaa servers</code> . Does the <code>automate-tester</code> user <code>pf-probe</code> get a response?

14 Rollback

Single port (fast)

```
default interface GigabitEthernet1/0/X
interface GigabitEthernet1/0/X
  switchport mode access
  switchport access vlan 99
```

Whole switch, keeping IBNS 2.0

```
default interface range GigabitEthernet1/0/1 - 46
interface range GigabitEthernet1/0/1 - 46
  switchport mode access
  switchport access vlan 99
! restore Gi1/0/45 AP trunk as in §10.4
```

Full restore (back to legacy mode)

```
copy flash:pre-dot1x.cfg startup-config
reload
```

PacketFence

```
tar xzf /root/pf-conf-<date>.tgz -C /
/usr/local/pf/bin/pfcmd configreload hard
```

OPNsense: *System* → *Configuration* → *History* → revert to the revision before the change, on both nodes.

Appendix C Changes in version 1.1

Corrections after the rollout at both sites. Each one cost time during the pilot, so they are listed with the symptom they caused.

Section	Change	Symptom it caused
\$4.2 Networks	Layer 2 networks always use PacketFence's own IP as the client gateway; the field only exists for routed networks	The gateway named in v1.0 could not be found in the UI
\$4.5 AD source	Realms <code>MFRACE</code> / <code>mfrace.internal</code> with <i>Strip on RADIUS authorization</i> , and <code>dnsHostName</code> in the search attributes	802.1X users and computers were rejected ("no role computed by any sources"), and every PC fell back to MAB and VLAN 555
\$4.6 RADIUS filter	Answer syntax is <code>Cisco-AVPair = ...</code> , not <code>=></code>	The switch received <code>"> interface-template-name=..."</code> and ignored it, so the AP port stayed an access port
\$10 AP template	<code>multi-host + port-control auto</code> ; templates merge rather than replace; no <code>switchport</code> commands on the interface; <code>force-authorized</code> causes a loop	The trunk stayed operationally down, Wi-Fi clients caused security violations, and one variant re-authenticated every 40 seconds
\$11 Windows	<i>Fallback to unauthorized network access</i> via a LAN profile, plus a note on <code>dot3svc</code> and on EAP-TLS vs PEAP	With PacketFence down the port was authorized in VLAN 20 but Windows blocked the connection; on another PC the stopped service meant no EAPOL at all
\$12 Isolation test	A CoA re-authenticates without bouncing the port, so Windows keeps its old address for up to about a minute; the built-in <i>Generic</i> event lets the user release themselves (set <i>Max enable</i> to 0)	The isolated PC appeared not to move to VLAN 666, then let itself out of isolation

Also added since v1.0: guest VLAN 100 (192.168.100.0/24 per site, internet only, kept apart in NetBox by the VRFs GUEST-DC1 / GUEST-DC2) and the switch standard `exec-timeout 15 0` on the console and all VTY lines.

Appendix A Other findings from the audit

These were found during the audit. They don't block the 802.1X rollout, but they're worth fixing.

#	Finding	Recommendation
1	SRV-PKI-01 (VM 110) is stopped	Check whether it's intentional. It could be an offline root CA. Certificate requests must go to SRV-PKI-02.
2	OPNsense DC1 alias <code>OD_ALL_VLANS</code> leaves out 10.10.10.0/24	Only affects the legacy <code>enc0</code> rules (<code>ipsec10</code> passes everything). Add it for consistency.
3	NW-DSW-DC2 uses the legacy <code>radius-server host 10.0.10.16</code> syntax. DC1 uses <code>radius server</code>	Align on the <code>radius server</code> block the next time you touch it.
4	NW-DSW-DC2 is VTP server with an empty domain. The DC1 DSW is transparent	Set <code>vtp mode transparent</code> for consistency and to avoid accidental VLAN changes.
5	DSWs run <code>spanning-tree mode pvst</code> , the ASWs run <code>rapid-pvst</code>	Move the DSWs to rapid-pvst in a maintenance window.
6	Repository configs (<code>Cisco/*/*.ios</code>) differ from the live devices (AAA, AP port, versions)	After the rollout, export the running configs to the repository. Remove the plaintext <code>cisco</code> passwords from the repository files.
7	PacketFence default realms point at domain <code>mfrace</code>	OK as it is. Users can sign in as <code>user</code> , <code>MFRACE\user</code> or <code>user@mfrace.internal</code> .
8	Management SSH to OPNsense and PacketFence uses the root account	Create personal admin accounts and turn off root login over SSH.

Appendix B Implementation checklist

✓	Task	Owner	Date
☐	Backups: PacketFence conf tarball, OPNsense config (both sites), <code>flash:pre-dot1x.cfg</code> on both ASWs		
☐	NetBox: 4 new prefixes, old prefixes deleted, VLAN name fixed		
☐	PacketFence: eth1/eth2 re-addressed, 4 networks created		
☐	PacketFence: switch group VLANs, CoA, types, sample entries deleted		
☐	PacketFence: role <code>unifi_ap</code> , AP nodes registered (DC1 + DC2)		
☐	PacketFence: RADIUS filter <code>UNIFI_AP_TEMPLATE</code>		
☐	PacketFence: AD rule, connection profile <code>wired-dot1x</code>		
☐	PacketFence: RADIUS certificate from AD CS		
☐	OPNsense DC1: VLANs + interfaces (both nodes), VIPs, aliases, rules, HA sync		
☐	OPNsense DC2: as DC1, plus DHCPRelay PF-REG / PF-JAIL		
☐	DSW DC1: Po45 + 555,666		
☐	DSW DC2: Po45, Po47, Po48 + 555,666		
☐	GPO <code>MFRACE-Wired</code> linked and tested on one PC		
☐	ASW DC1: IBNS 2.0, global config, templates, pilot port		
☐	Pilot tests §12.2 passed (DC1)		
☐	ASW DC1: rollout Gi1/0/1-46, PacketFence mode Production		
☐	ASW DC2: same as DC1, pilot tests passed		
☐	Repository updated with the new running configs		